

Rachel Marston

Interdisciplinary social science researcher interested in the intersections of environmental and climate justice, community-led solutions, and anti-colonial frameworks.

rcmarst4@gmail.com | rachelmarston.github.io

Education

Tufts University

Boston, MA, USA

B.A., Applied Environmental Studies and Anthropology

2017-2021

Advisor: Ninian Stein

Senior Honors Thesis: *Unsettling Wright-Locke and Re-Storying the Land: Centering Indigenous Histories in Environmental Education*

Honors: Graduated *summa cum laude*, Highest Honors in Senior Thesis, Phi Beta Kappa

Work and Research Experience

Research Project Manager, External Capacity Building; Environmental Defense Fund 2023 — present

- Conducts mixed-methods research involving 30+ interviewees and 82 survey respondents to provide clear recommendations for organizational strategy and program implementation; translate key research findings into deliverables for internal and external audiences, including 2 presentations to donors, 3 conference presentations, 6 memos, and 2 academic manuscripts.
- Leads strategy development and program execution for the External Capacity Building pillar of EDF's Justice and Equity strategic plan, including the implementation of the Frontline Resource Institute (FRI) project and related research strategy.
- Designs and manages resource sharing for the Frontline Resource Institute's external resource hub, including conceptualization, research, design, and distribution of 4 quarterly newsletters, 12 monthly resource sharing emails, and a blog series; facilitated network growth from 366 to 481 in 2024.
- Facilitates and plans quarterly meetings for FRI's Conceptual Committee—composed of 11 environmental and climate justice leaders—to guide the project, including development and approval of FRI's project strategic plan; led document design and distribution strategy to 400+ community-based organizations and EDF stakeholders.

Community Capacity Center Tom Graff Fellow, Environmental Defense Fund

2021-2023

- Tracked and managed goals and project implementation for the Frontline Resource Institute pilot project.
- Conducted qualitative and quantitative research—including interviews and surveys—to understand the needs and priorities of frontline, environmental justice community-based organizations.
- Distributed \$1,000,000 in philanthropic funds through the Frontline Resource Institute's inaugural grant cycle, managing an application process that garnered 314 applicants.

Papers in Preparation

[IP1] [In Prep] *Supporting Frontline Communities through Intermediary Organizations: A Case Study of the Frontline Resource Institute*. R. Marston, N. Lutz, G. Ainsa, D. Mangabat, J. Stober, M. Brown, K. Turner.

Peer Reviewed Publications

Journal Papers (Heavily Reviewed, Archival):

* = equal contribution/co-author

[J4] [Under Review] *Lemonade from Lemons: AI Practices of Use, Refusal, and Innovation in Frontline Communities*. K. Turner, **R. Marston**, N. Lutz, M. Brown.

[J3] [Under Review] *AI on the Frontlines: A Framework to Explore the Environmental Justice Impacts of Emergent Technology Systems*. K. Turner, **R. Marston**, A. Dogan, N. Lutz, M. Brown. 2026. Journal of Responsible Computing.

[J2] [Under Review] "What Is Our Strategy In Light of This?": *Frontline Narrative and Storytelling in the Sociodigital Era*. N. Lutz*, **R. Marston***, J. Camuzeaux, M. Brown, K. Turner. Community Science. 2026. [* = co-first author]

[J1] *A mixed-methods needs assessment of frontline communities: Insights for engagement and partnerships between communities and intermediary organizations*. Community Science, 4, e2025CSJ000133.

<https://doi.org/10.1029/2025CSJ000133>. **R. Marston**, N. Lutz, G. Ainsa, D. Mangabat, J. Stober, M. Brown, K. Turner. (2025)

Abstracts, Presentations, and Posters (Lightly Reviewed, Semi-Archival):

[P4] *AI and Frontline Communities*. AGU Fall Meetings Abstracts. N. Lutz, **R. Marston**, J. Stober, M. Brown, K. Turner. (2025)

[P3] *Narrative and storytelling sovereignty for frontline communities: ownership, autonomy, and support in a 21st century context*. AGU Fall Meetings Abstracts. **Rachel Marston**, Nina Lutz, Jonathan Camuzeaux, Margot Brown, Katlyn M. Turner. (2025)

[P2] *A mixed-methods systems analysis of an intergenerational intermediary organization: insights on equitable resource provision for frontline communities*. AGU Fall Meeting Abstracts 2024. **R. Marston**, N. Lutz, G. Ainsa, D. Mangabat, J. Stober, M. Brown, K. Turner. (2024)

<https://ui.adsabs.harvard.edu/abs/2024AGUFMSY11C2545M/abstract>

[P1] *Understanding the Needs and Priorities of Frontline Communities Using a Co-creative Systems Engineering Approach*. AGU Fall Meeting Abstracts 2023. **R. Marston**, N. Lutz, G. Ainsa, D. Mangabat, J. Stober, M. Brown, K. Turner. (2023) <https://ui.adsabs.harvard.edu/abs/2023AGUFMGC33D11.9M/abstract>

Workshop Papers (Lightly Reviewed, Semi-Archival):

[W1] *Frontline-Grounded AI Futures: Supporting Environmental and Climate Justice Communities Navigating AI Tools, Technologies, and Infrastructures*. HCI-Terra CHI 2026 Workshop. **R. Marston**, **N. Lutz**, M. Brown, K. Turner. CHI 2026.

https://hci-terra.github.io/assets/submissions/CHI_2026_HCI_TERRA_FrontlineCommunities_AI.pdf (2026)

Conference Convenings

Frontline Community-Driven Approaches to Climate and Environmental Change: Theory, Narrative, and Practice. **American Geophysical Union, Fall Meeting**. December 18, 2025

Storytelling and Narrative Sovereignty in Frontline Communities. **Center for Engagement, Environmental Justice, and Health Annual Symposium.** September 12, 2025

Building Organizational Capacity and Durability: The Frontline Resource Institute. **Center for Engagement, Environmental Justice, and Health Annual Symposium.** September 12, 2023

Teaching Experience

Part-Time Instructor, Tufts University Department of Environmental Studies **Fall 2025**

Course: ENV150: Environment, Communication, and Culture

Teaching Assistant, Tufts University Department of Environmental Studies **Fall 2020**

Course: ENV1: Introduction to Environmental Studies | Professor: Dr. Ninian Stein

Public Scholarship (Internally Reviewed)

[O1] [Best Practices for Frontline Narrative and Storytelling Sovereignty: A Guide](#). KTLN Consulting and Frontline Resource Institute at Environmental Defense Fund. **R. Marston**, N. Lutz, K. Turner. (2025)

Selected Press

EOS Magazine, [Rethinking Engagement with Frontline Communities](#) **Jan 2026**

Invited Talks, Guest Lectures, and Panels

Guest Lectures and Talks

Tufts University, [Hoch Cunningham Environmental Lectures](#) **Jan 2025**

Tufts University Environmental Studies Department

Course: ENV150: Environment, Communication, and Culture **Jan 2025**

Course: ENV1: Introduction to Environmental Studies **Feb 2025**

Panels

Cultivating Leadership for Equitable and Just Systems Change, EDF All-Staff Retreat **May 2023**

Current Professional Societies

American Geophysical Union ([AGU](#))

Special Interest Group on Human Computer Interaction ([SIGCHI](#))

Service

Environmental Defense Fund, Climate Corps Environmental and Climate Justice Cohort Advisor (2022-2025)